

# Algebraic Techniques

## Week 1D

# Overview

## *Factorisation*

- Key Ideas*
- *Monic Quadratic Factorisation*
  - *Non-Monic Quadratic Factorisation*
  - *Combined Factorisation*

## Key Idea - Quadratic Factors - Monic Quadratic Trinomials

An algebraic expression of the form  $x^2 + bx + c$  is called a *monic quadratic trinomial*.

Monic  $\Rightarrow$  coefficient of  $x^2$  is 1

Quadratic  $\Rightarrow$  highest power of  $x$  is 2

Trinomial  $\Rightarrow$  expression contains 3 terms

### Discussion Question:

Consider the following expansion:

$$\begin{aligned}(x+2)(x+3) &= x^2 + 3x + 2x + 6 \\ &= x^2 + 5x + 6\end{aligned}$$

How could we factorise this quadratic expression without being given the above expansion?

What do you notice about the coefficient of  $x$  and the constant term on the right-hand side, and the numbers in the binomial on the left hand side?

$$2 + 3 = 5$$

$$2 \times 3 = 6$$

There are several methods for factorising quadratic trinomials. One method is given here:

$$\begin{aligned}(x + a)(x + b) &= x(x + b) + a(x + b) \\ &= x^2 + xb + ax + ab \\ &= x^2 + (a + b)x + ab\end{aligned}$$

The *important thing to notice* is that,

- the *coefficient of  $x$*  in the result is the *sum  $a + b$* .

and

- the *constant* in the result is the *product of  $a$  and  $b$* .

So, we look for two numbers  $a$  and  $b$  whose sum is the coefficients of  $x$  (which equals  $b$  in the general quadratic expression  $x^2 + bx + c$ ) and whose product is the constant term (which equals  $c$  in the general quadratic expression). Once the values of  $a$  and  $b$  have been established, the factors can be easily obtained.

### *Examples:*

Factorise  $x^2 + 5x - 6$

Here we have a quadratic expression. We need two numbers whose ***product is -6***.

Possible pairs of numbers are:  $1 \times -6$ ,  $-1 \times 6$ ,  $2 \times -3$ ,  $-2 \times 3$

Of these pairs, we need one pair whose ***sum is +5***. These two factors must be **-1** and **6**.

Hence  $x^2 + 5x - 6 = (x - 1)(x + 6)$ .

Factorise  $y^2 - 8y + 15$

Here we have a quadratic expression. We need two numbers whose ***product is 15***.

Possible pairs of numbers are:  $1 \times 15$ ,  $3 \times 5$ ,  $-1 \times -15$  and  $-3 \times -5$ .

Of these pairs, we need one pair whose ***sum is -8***. These two factors must be **-3** and **-5**.

Hence  $x^2 - 8x + 15 = (x - 3)(x - 5)$ .

**Exercise:** Factorise the following expression:  $x^2 - 3x - 40$

↑ ↑  
two numbers  
that add to  
give  $-3$

$$a + b = -3$$

$$-8 + 5 = -3$$

call the 2 numbers  
a and b  
two numbers  
that mult to  
give  $-40$

$$a \times b = -40$$

$$-8 \times 5 = -40$$

$$\therefore (x^2 - 3x - 40) = (x + 5)(x - 8)$$

## Key Idea - Quadratic Factors - Monic Quadratic Trinomials

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An algebraic expression of the form

$$ax^2 + bx + c$$

is called a ***non-monic quadratic trinomial***, where  $a$  is any number except 1 or 0.

Factorising a non-monic trinomial is a bit more complicated. So, the first thing to do is see if you can take the leading coefficient out of the whole quadratic trinomial as a common factor, and then just factorise the remaining monic quadratic trinomial.

### ***Example:***

Factorise  $3x^2 + 21x + 18$ .

Here the leading coefficient of 3 is a common factor, so  $3x^2 + 21x + 18 = 3(x^2 + 7x + 6)$ .

We notice that  $1 + 6 = 7$  and  $1 \times 6 = 6$ , and so  $x^2 + 7x + 6 = (x + 1)(x + 6)$ .

$$3x^2 + 21x + 18 = 3(x + 1)(x + 6)$$

If you like, you can rewrite this as  $(3x + 3)(x + 6)$ , but the other form is probably more useful.

Of course, you have to be pretty lucky to be able to take out the leading coefficient as a common factor. Otherwise, we just have to work a bit harder. In these notes we have included two different methods of factorizing non-monic quadratic trinomials.

### Method 1:

The method we are going to use to factorise non-monic quadratic trinomials is an extension of the method we have used for monic quadratic trinomials.

Consider factorising  $3x^2 + 2x - 8$ .

1. Firstly, find the product of the first and third terms and use this result to decide how to rewrite the middle term:
2. We now need two factors whose product is  $-24x^2$ . Possible pairs of factors are:  
 $6x \times -4x, 12x \times -2x, 8x \times -3x, -6x \times 4x, -12x \times 2x, -8x \times 3x$
3. Of these pairs of factors, we need one pair whose sum is  $2x$ . These two factors are  **$+6x$  and  $-4x$** . Rewrite by replacing  $2x$  with  $(6x - 4x)$ :  
$$3x^2 + 2x - 8 = 3x^2 + 6x - 4x - 8$$
4. Factorise by grouping in pairs:  
$$3x^2 + 2x - 8 = 3x(x + 2) - 4(x + 2)$$
$$= (3x - 4)(x + 2)$$



***Note:***

*Some quadratic expressions cannot be factorised.*

Not every quadratic trinomial can be factorised, and even among those that can, the factors might not be integers. We won't go into the details in this subject, but for future reference:

The quadratic trinomial

$$ax^2 + bx + c$$

can be factorised if and only if

$$b^2 - 4ac \geq 0.$$

## Method 2:

If we are given a quadratic trinomial in the form

$$Ex^2 + Fx + G$$

and we want to factorise it, we start by considering what happens if we multiply two arbitrary linear polynomials:

$$\begin{aligned}(kx + l)(mx + n) &= kx(mx + n) + l(mx + n) \\ &= (km)x^2 + knx + lmx + ln \\ &= (km)x^2 + (kn + lm)x + ln\end{aligned}$$

The coefficient of  $x^2$  is  $km$ , the coefficient of  $x$  is  $(kn + lm)$  and the constant coefficient is  $ln$ . So, we are looking for numbers  $k, l, m, n$  such that

$$km = E$$

$$kn + lm = F, \text{ and } ln = G$$

If we can find numbers like that, we have found a factorisation of the original quadratic trinomial.

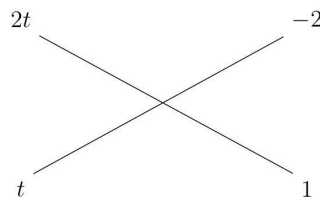
### Example:

Factorise  $2t^2 + 3t - 2$ .

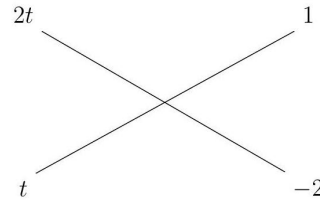
Since the leading coefficient 2 is not a common factor, we look for numbers  $k, l, m, n$  such that  $km = 2, kn + lm = 3$  and  $ln = -2$ . There aren't many factors of 2 and -2 :  $km = 2$ . We can try  $k = 1, m = 2$  or  $k = 2, m = 1$  or  $k = -1, m = -2$  or  $k = -2, m = -1$ . Likewise, there aren't many factors of -2 . We can try  $l = 1, m = -2$  or  $l = 2, m = -1$  or  $l = -1, m = 2$  or  $l = -2, m = 1$ .

We can just start trying combinations, but it might be worth noticing first that with all of these numbers the values of  $kn$  and  $lm$  are 1, 2, 4, -1, -2 or -4 . We want them to add up to 3 , so we're probably looking for  $2 + 1$  or  $4 + -1$ .

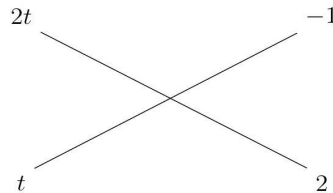
Now try some pairs. It can help to arrange them in a cross where the two terms on the left multiply to give  $2t^2$  and the two on the right multiply to give  $-2$  like this:



Add up the product at opposite ends of the diagonal lines and see if they sum to  $3t$ . In this instance, we get  $2t - 2t = 0 \neq 3t$ , so we try again.



Summing the products on the diagonal, we get  $-4t + t = -3t \neq 3t$ , so we try again, but this gives us a hint: if we change the signs of both numbers on the right, we won't change their product, but we will reverse the sign of the sum. So, we try



Now summing the products on the diagonal we get  $4t - t = 3t$  as required. So we have found our factorisation. We read off the factors from the rows of our cross diagram. The first row corresponds to  $2t - 1$  and the second to  $t + 2$ , so we get the factorisation

$$2t^2 + 3t - 2 = (2t - 1)(t + 2) \checkmark$$

You should always double-check your work by expanding the right-hand side. Here

$$(2t - 1)(t + 2) = 2t^2 + 4t - t - 2 = 2t^2 + 3t - 2. \checkmark$$

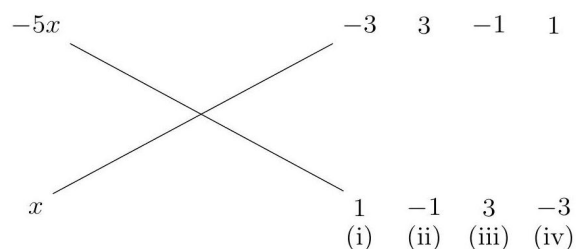
*Examples:*

Factorise  $16x - 3 - 5x^2$ .

Step one is to rewrite it in the usual order:  $-5x^2 + 16x - 3$ .

We want numbers  $k, l, m, n$  so that  $km = -5$  and  $ln = -3$  and  $kn + lm = 16$ . Neither  $-5$  nor  $-3$  has many factors. Either  $k = -5, m = 1$  or  $k = 5, m = -1$ . And likewise,  $l, n$  are  $\pm 1$  and  $\pm 3$  with opposite signs.

Let's start with all the possibilities where  $k = -5$  and  $m = 1$  :



We check the sums of the products along the diagonal in terms of  $\gamma$ .

For (i):  $-5x - 3x = -8x \neq 16x$ .

You could keep just going through all the options, but my reaction is that to get  $16x$  we need to be multiplying the  $-5x$  by  $-3$ , so I skip straight to trying (iv).

For (iv):  $15x + x = 16x$ . ✓

So, reading off the horizontal rows, we have our factorisation:

$$16x - 3 - 5x^2 = (-5x + 1)(x - 3)$$

**Exercises:** Factorise the following expression  $4x^2 - 12x + 9$ .

$$4x^2 - 6x - 6x + 9$$

note  
- x - = +

$$2x(2x - 3) - 3(2x - 3)$$

$$= (2x - 3)(2x - 3)$$
$$= (2x - 3)^2$$

you can  
leave the  
answer here

$$4x^2 \times 9 = 36x^2$$

$$a \times b = 36x^2$$

$$a + b = -12x$$

$$\therefore a = -6x$$

$$b = -6x$$

## Key Idea - Combining Methods of Factorising

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In some cases, it is necessary to use more than one method to factorise the expression.

### *Examples:*

Factorise fully:

$$\begin{aligned} 3x^2 - 3y^2 &= 3(x^2 - y^2) \\ &= 3(x - y)(x + y) \end{aligned}$$

$$\begin{aligned} 64 - 36y^2 &= 4(16 - 9y^2) \\ &= 4(4^2 - (3y)^2) \\ &= 4(4 - 3y)(4 + 3y) \end{aligned}$$

$$\begin{aligned} 4x^2 - 4x - 48 &= 4(x^2 - x - 12) \\ &= 4(x + 3)(x - 4) \end{aligned}$$

$$\begin{aligned} 3x^4 - 48 &= 3(x^4 - 16) \\ &= 3((x^2)^2 - 4^2) \\ &= 3(x^2 - 4)(x^2 + 4) \\ &= 3(x^2 + 4)(x - 2)(x + 2) \end{aligned}$$

**Exercises:** Factorise the following:

(a)  $6x^3 + 3x^2 - 3x$   $= 3x(2x^2 + x - 1)$  common factor of  $3x$   
 $= 3x(2x^2 + 2x - x - 1)$  non-monic  
 $= 3x(2x(x+1) - (x+1))$  Quadratic  
 $= 3x(2x-1)(x+1)$

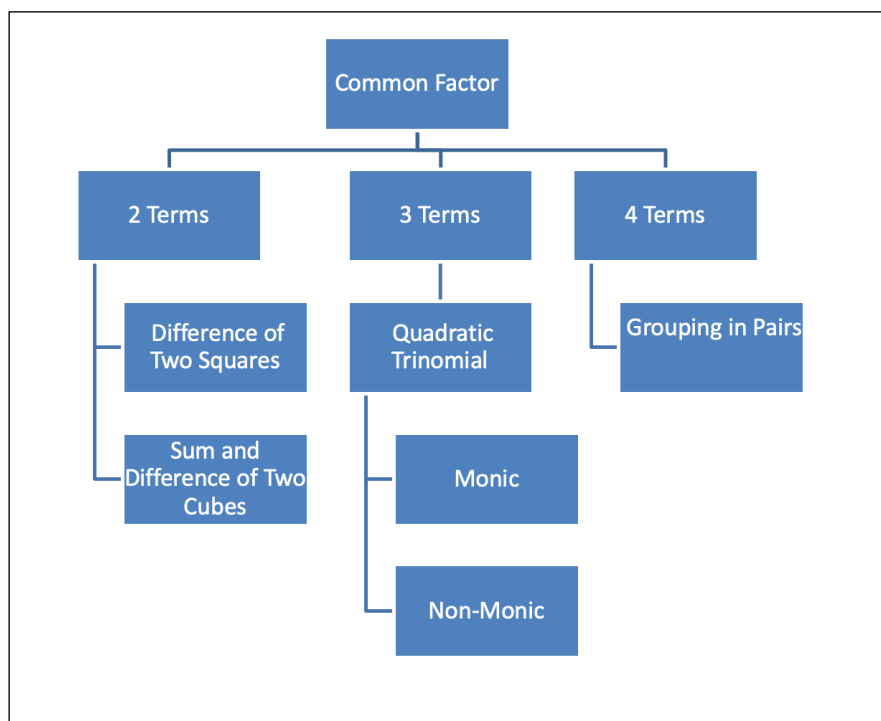
(b)  $8x^3 + 27$

$$(2x)^3 + 3^3 = (2x + 3)((2x)^2 - 2x \times 3 + 3^2)$$
$$= (2x + 3)(4x^2 - 6x + 9)$$



## Mixed Factorisation: Summary

When asked to factorise an expression, you need to decide on the strategies to use. The following flow chart will help guide you in this process, using the factorisation techniques that are presented in the lectures.



Steps:

1. Always look for any **simple** first (take out the Highest Common Factor).
2. If there are **two terms**, look for a **difference of two squares** or a **sum or difference of two cubes**.
3. If there are **three terms**, look for a **quadratic trinomial**. Is it a **monic** or **non-monic trinomial**?
4. If there are **four terms**, look for **grouping in pairs**.

***Remember to check:*** That you cannot factorise further.